

Improving transferability of SDM

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Model transferability

Ability of a model to predict beyond the data it was originally trained on.

Particularly **relevant when forecasting species distributions** under **climate change**.

Mainly dependent on:

1. Adequacy of environmental and species occurrence data to train the models;
2. Reduced complexity of the models (e.g., number of variables);
- 2. Proper choice / tuning of model parameterisation.**

Machine learning algorithms

Construct a function that estimates the effect of different environmental variables on the distribution of a species.

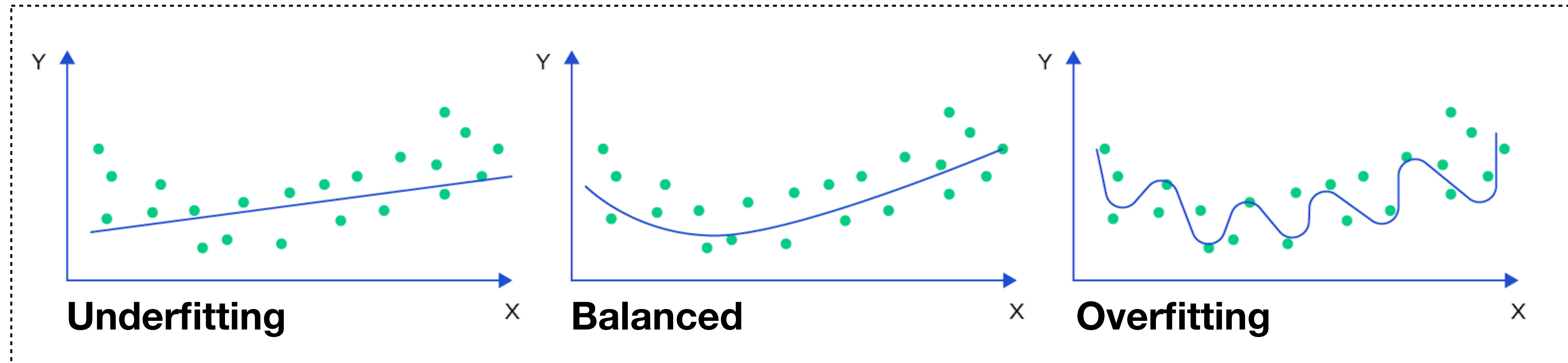
e.g., Maximum Entropy; Boosted Regression Trees; Extreme Gradient Boosting.

Advantages: (1) very high predictive performance; (2) can use continuous and categorical variables; (2) automatically detect interactions between variables**;
(3) **includes specific parameters that protect against overfitting;**

Major limitations: susceptible to overfitting if not properly parameterized.

*** when a variable has a different effect on the outcome depending on the values of another variable;*

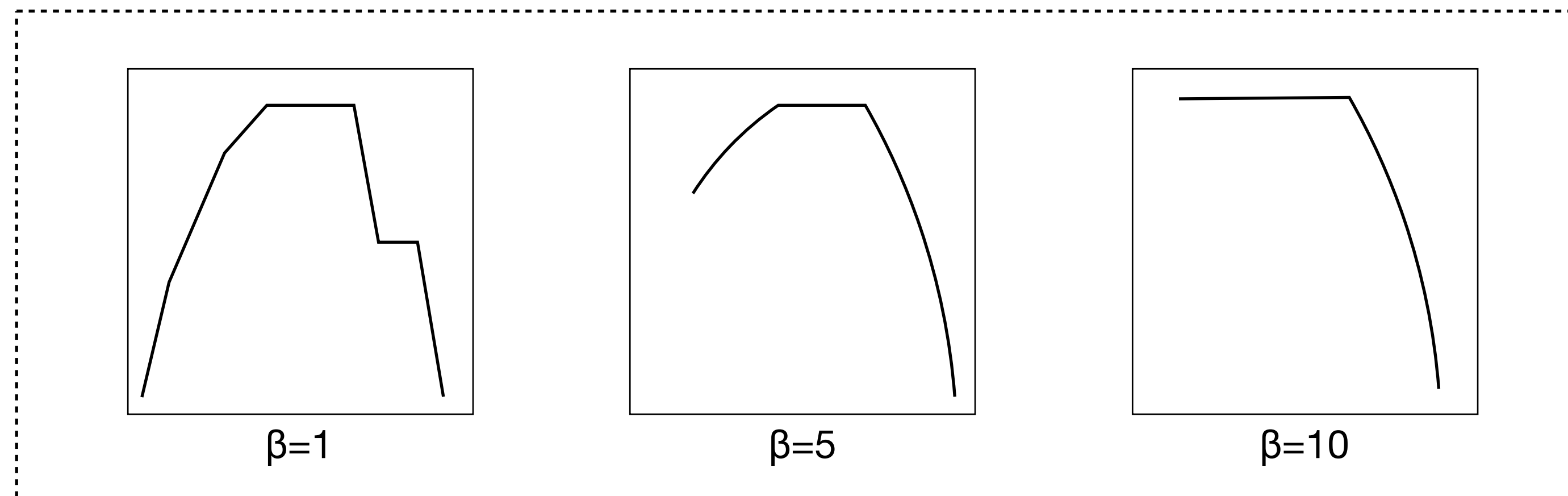
e.g., species' tolerance to oxygen often depends on temperature; increased metabolic oxygen demand.



There is a set of optimal hyperparameters that best control the learning process (optimal hyperparameters will solve the learning problem; will return the best fit).

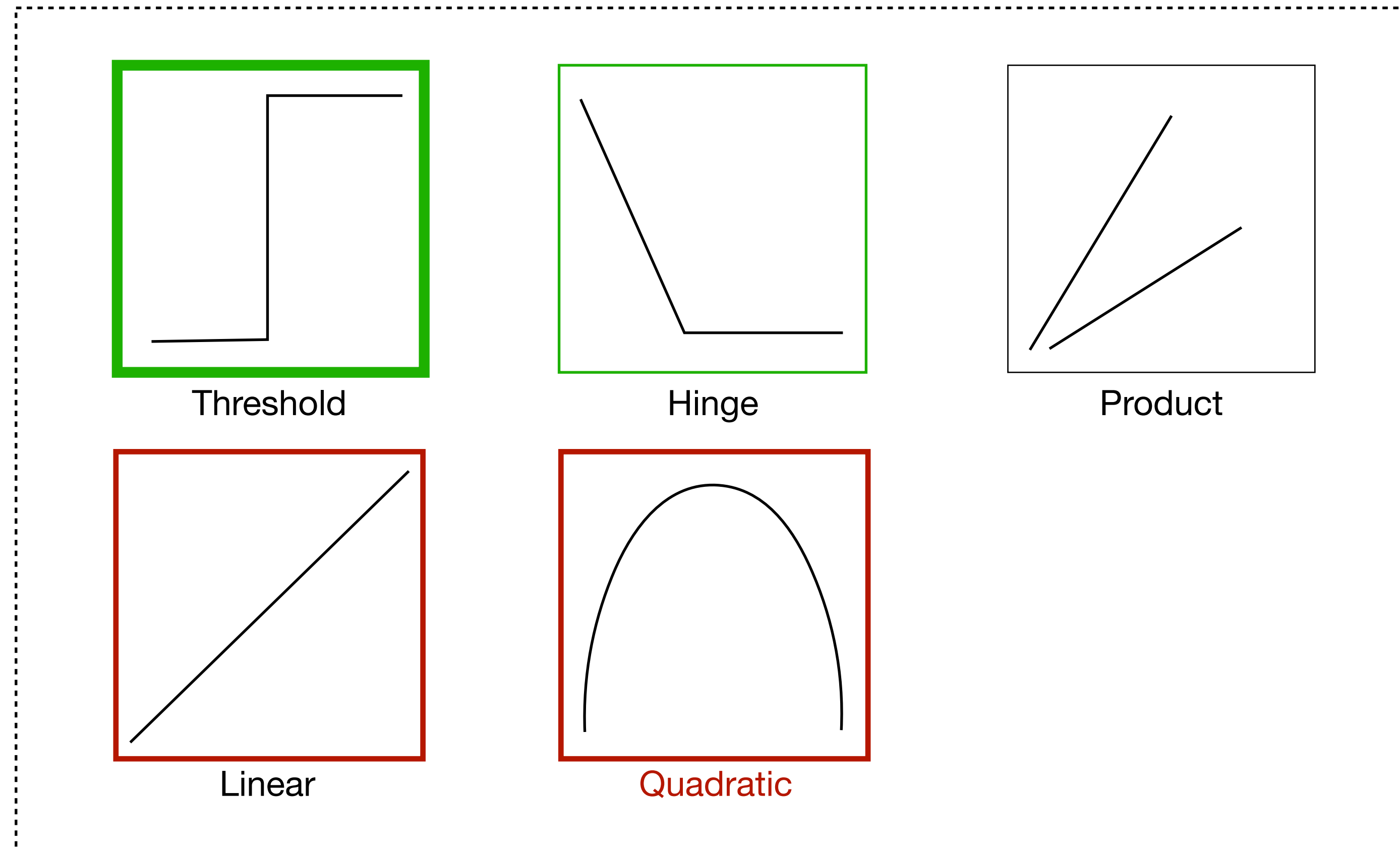
Maxent hyperparameters

A key parameter of **Maxent** is **regularization**, which controls **fit constraints** and therefore strongly impacts **overfitting**.



Higher regularization (β multiplier) values reduced the smoothing / relaxation of curves, which in term, can reduce overfitting.

To much regularization can lead to underfitting.



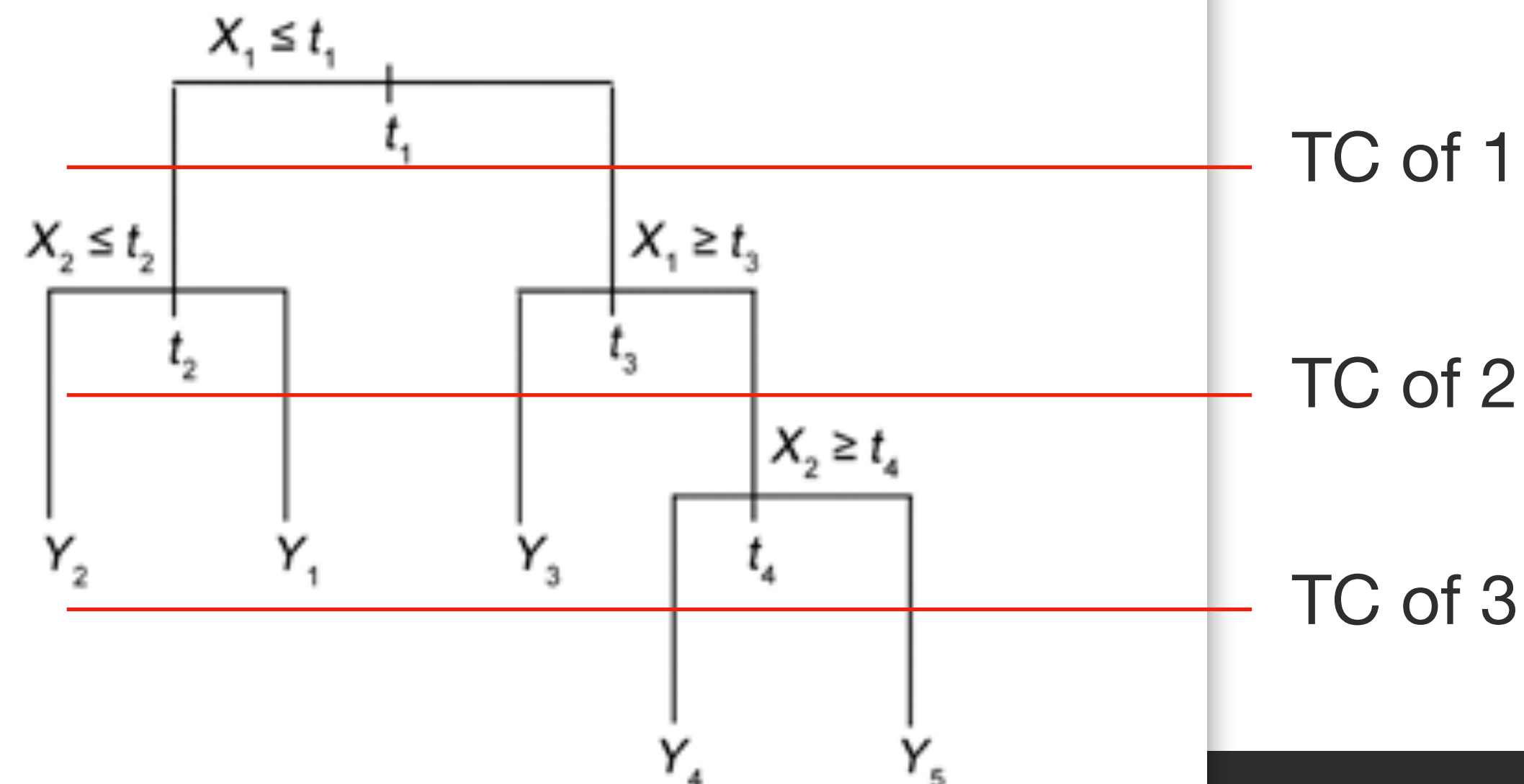
An additional parameter of Maxent is feature type.

We tend to **exclude** features that are more prone to lead to **overfitting** or to **underfitting**.

Boosted Regression Trees hyperparameters

Tree complexity: a TC of 1 (single decision; two terminal nodes), a TC of 2 fits a model with up to two-way interactions, and so on. Higher / Lower TC values control complexity, which can translate into overfitting or underfitting.

Learning rate: determines the **contribution of each tree to the final model**. Higher / Lower values control complexity, which can translate into overfitting or underfitting.



Hyperparameter optimization

There is no rule of thumb for the hyperparameters; the same algorithm may require different parameter values to generalize different datasets.

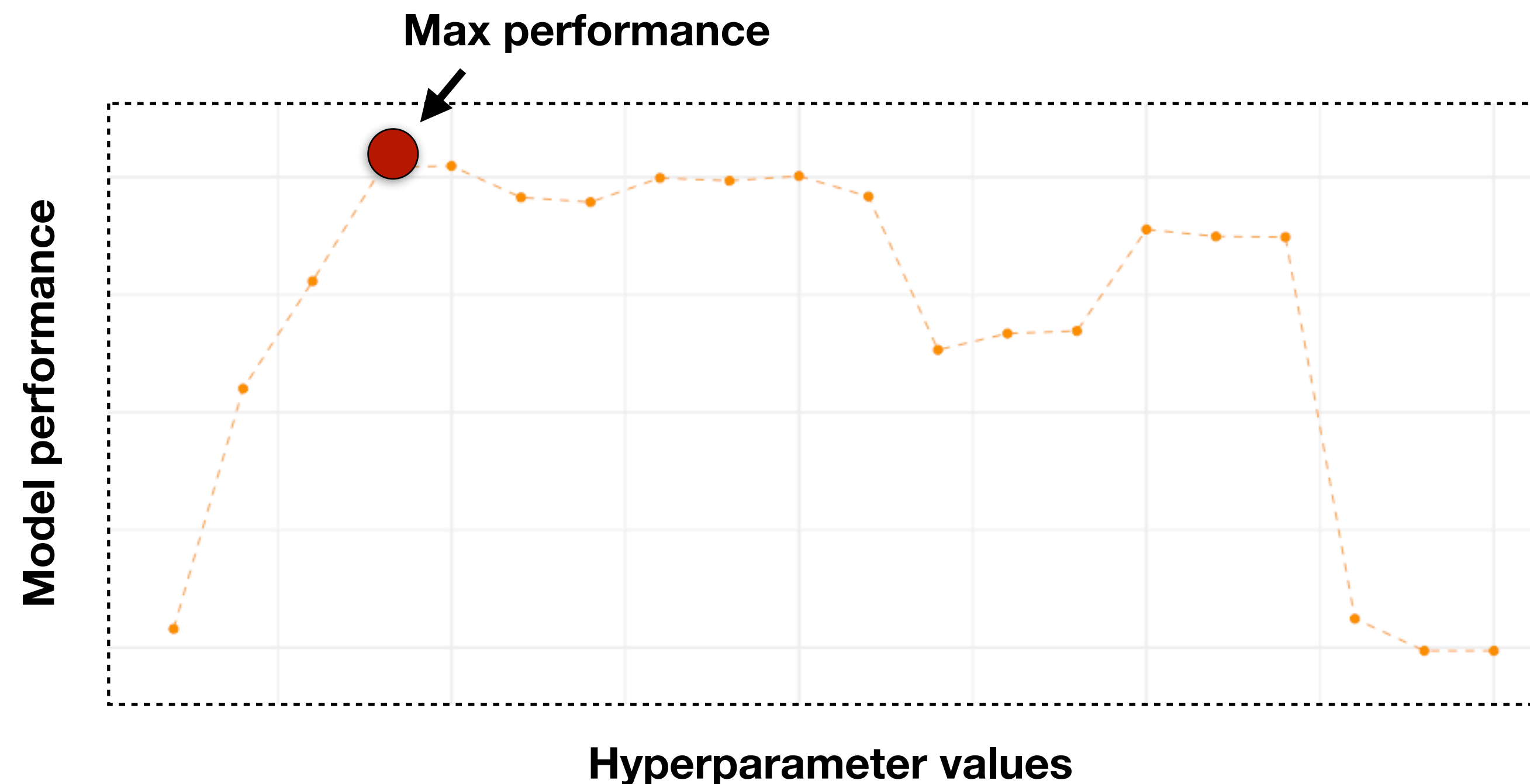
How to tune these parameters to improve model fitting?

Hyperparameter optimization

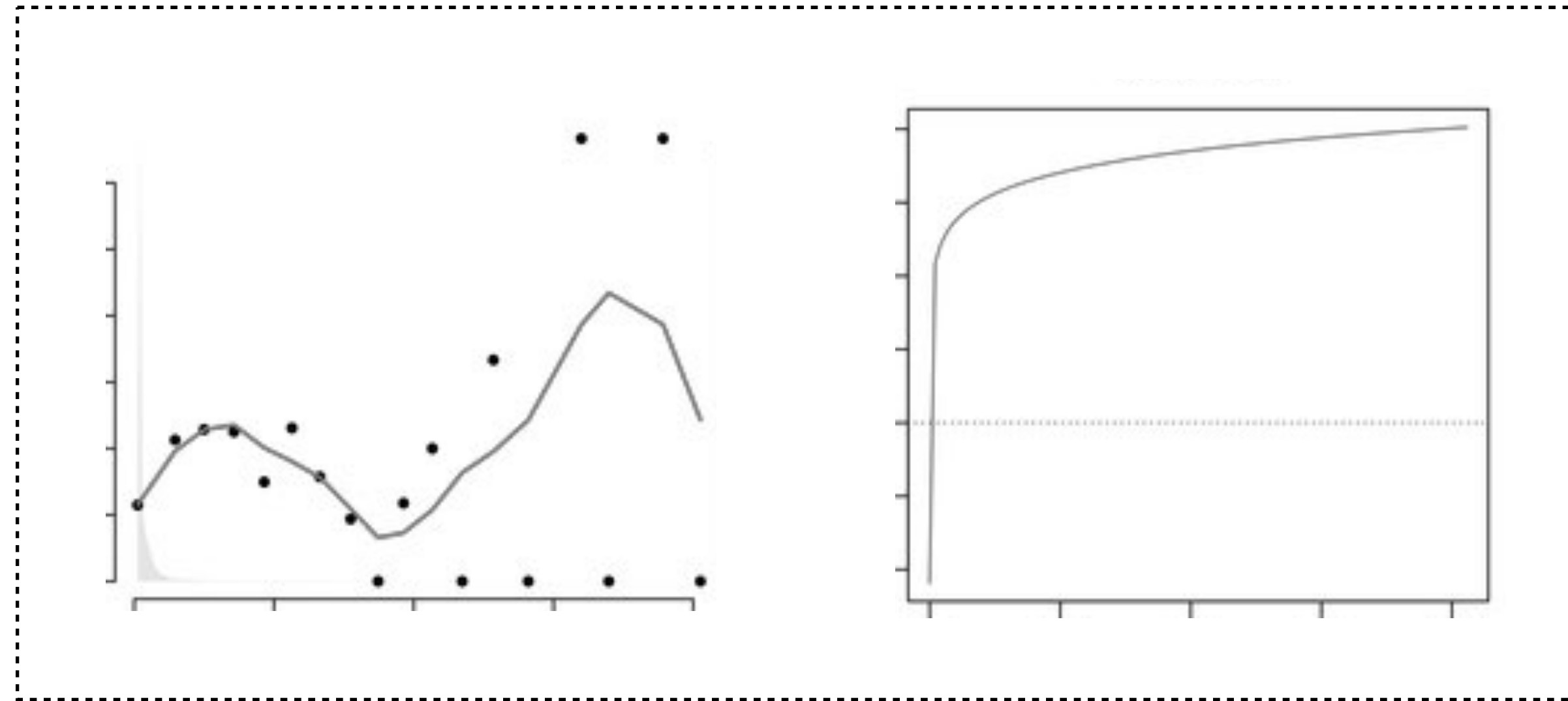
The approach relies on testing a span of parameter values and finding which value maximizes the performance of models.

Cross-validation (tested in validation data) is used in the process.

Which parameter value has higher performance in independent data?



Overfitted response vs. positive monotonicity



Guiding the model: using monotonicity constraints

A way to force the modeled relationship to follow a pre-defined, simple trend.

This prevents the model from fitting overly complex, potentially unrealistic shapes to the data for that variable.

Only a few machine learning algorithms have their implementation (BRT; XGBoost)

The choice of monotonicity constraints is made a priori per variable **based on ecological theory** (no monotonicity, negative or positive monotonicity).